# Ziyi (Zoe) Wang

Ph.D. Student, Department of Computer Science & Engineering, Texas A&M University
Research: Human-Centered AI · Human–AI Interaction · Natural Language Processing
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## Education

2026 – **Ph.D. in Computer Science** — Texas A&M University
Advisor: Prof. Meng Xia
Expected graduation: May 2031

2024 – 2026 **M.S. in Human-Computer Interaction** — University of Maryland, College Park
Research advisors: Prof. Fumeng Yang, Dr. Zijian Ding
GPA: 4.00/4.00

## Publications

My research appears at top-tier HCI, NLP, and AI conferences (e.g., EMNLP, DIS, IEEE VIS, AAAI). Within each category, papers are listed newest first; * denotes equal contribution, and my name is highlighted. Google Scholar (October 2026): 159 citations, h-index 7.

### Peer-Reviewed Conference & Journal Papers

2026 P9 **Do Vision Language Models Understand Human Engagement in Games?**
Ziyi Wang*, Qizan Guo*, Rishitosh Singh*, Xiyang Hu
Conference on Empirical Methods in Natural Language Processing (EMNLP)
Oral Presentation (top 2.6%)

2026 P8 **StreetDesignAI: Broadening Designer Perspectives Through Multi-Persona Evaluation of Cycling Infrastructure**
Ziyi Wang, Yilong Dai, Duanya Lyu, Mateo Nader, Sihan Chen, Wanghao Ye, Zijian Ding, Xiang Yan
Proceedings of the ACM Designing Interactive Systems Conference (DIS)

2026 P7 **Made to Feel: How Designers Bring Emotions into Affective Visualization**
Yixin Bai, Ziyi Wang, Keke Wu, Fumeng Yang
IEEE Visualization and Visual Analytics Conference (VIS) | Short Papers
Best Paper Award (top 1%)

2026 P6 **Physics-Preserving Latent Compression for Zero-Shot Resolution Transfer in 3D Turbulence**
Yilong Dai, Yiming Sun, Yiheng Chen, Ziyi Wang, Shengyu Chen, Xiaowei Jia, Runlong Yu
IEEE International Conference on Data Mining (ICDM)

2026 P5 **Learning PDE Solvers with Physics and Data: A Unifying View of Physics-Informed Neural Networks and Neural Operators**
Yilong Dai, Shengyu Chen, Ziyi Wang, Xiaowei Jia, Yiqun Xie, Vipin Kumar, Runlong Yu
International Joint Conference on Artificial Intelligence (IJCAI) | Survey Track

2026 P4 **Mitigating Hallucinations in Large Language Models via Causal Reasoning**
Yuangang Li*, Yiqing Shen*, Yi Nian, Jiechao Gao, Ziyi Wang, Chenxiao Yu, Shawn Li, Jie Wang, Xiyang Hu, Yue Zhao
AAAI Conference on Artificial Intelligence (AAAI)

2025 P3 **Frontend Diffusion: Empowering Self-Representation of Junior Researchers and Designers Through Multi-agent System**
Zijian Ding, Qinshi Zhang, Mohan Chi, Ziyi Wang
IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)

2025 P2 **JailDAM: Jailbreak Detection with Adaptive Memory for Vision-Language Model**
Yi Nian*, Shenzhe Zhu*, Yuehan Qin, Li Li, Ziyi Wang, Chaowei Xiao, Yue Zhao
Conference on Language Modeling (COLM)

2025 P1 **Few-Shot Graph Out-of-Distribution Detection with LLMs**
Haoyan Xu*, Zhengtao Yao*, Yushun Dong, Ziyi Wang, Ryan A. Rossi, Mengyuan Li, Yue Zhao
European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD)

### Peer-Reviewed Workshop Papers

2026 W2 **Multimodal Generative Engine Optimization: Rank Manipulation for Vision-Language Model Rankers**
Yixuan Du, Chenxiao Yu, Haoyan Xu, Ziyi Wang, Yue Zhao, Xiyang Hu
Workshop on Towards Knowledgeable Foundation Models (KnowFM) at ACL

2025 W1 **From Image Generation to Infrastructure Design: A Multi-agent Pipeline for Street Design Generation**
Chenguang Wang, Xiang Yan, Yilong Dai, Ziyi Wang, Susu Xu
Workshop on Urban AI (UrbanAI) at NeurIPS

### Preprints

2026 A7 **Persona-aware and Explainable Bikeability Assessment: A Vision-Language Model Approach**
Yilong Dai*, Ziyi Wang*, Chenguang Wang, Kexin Zhou, Yiheng Qian, Susu Xu, Xiang Yan
arXiv preprint

2026 A6 **ResearchCube: Multi-Dimensional Trade-off Exploration for Research Ideation**
Zijian Ding, Fenghai Li, Ziyi Wang, Joel Chan
arXiv preprint

2026 A5 **Sovereign-OS: A Charter-Governed Operating System for Autonomous AI Agents with Verifiable Fiscal Discipline**
Aojie Yuan, Haiyue Zhang, Ziyi Wang, Yue Zhao
arXiv preprint | demo paper | GitHub (95 stars)

2026 A4 **In the Shadow of Judgment: Mapping Out the Landscape of Human–AI Decision-Making Through a Systematic Review**
Yixin Bai, Taehyun Yang, Zhongzheng Xu, Dayeon Ki, Ziyi Wang, Yu Hou, Fumeng Yang
Preprint (HAL)

2025 A3 **CareerPooler: AI-Powered Metaphorical Pool Simulation Improves Experience and Outcomes in Career Exploration**
Ziyi Wang, Ziwen Zeng, Yuan Li, Zijian Ding
arXiv preprint

2025 A2 **Graph Synthetic Out-of-Distribution Exposure with Large Language Models**
Haoyan Xu*, Zhengtao Yao*, Ziyi Wang, Zhan Cheng, Xiyang Hu, Mengyuan Li, Yue Zhao
arXiv preprint

2025 A1 **GLIP-OOD: Zero-Shot Graph OOD Detection with Graph Foundation Model**
Haoyan Xu*, Zhengtao Yao*, Xuzhi Zhang, Ziyi Wang, Langzhou He, Yushun Dong, Philip S. Yu, Mengyuan Li, Yue Zhao
arXiv preprint

## Research Experience

2026 – **Research Assistant** — Dream Lab, Texas A&M University
Advisor: Prof. Meng Xia
Ph.D. research on how AI can simulate people: building and evaluating simulations that range from role-playing personas with different needs to predicting human judgments and latent states such as engagement and affect, and exploring how they can help people understand others and make better decisions.

2024 – 2026 **Research Assistant** — HCIL, University of Maryland, College Park
Advisors: Prof. Fumeng Yang, Dr. Zijian Ding
Led one project (A3) on generative-AI career exploration and contributed to four projects (P7, P3, A6, A4) on affective visualization, agentic design tools, research ideation, and human–AI decision-making.

2025 – 2026 **Research Assistant** FORTIS Lab, University of Southern California
Advisor: Prof. Yue Zhao
Contributed to six projects (P4, P2, P1, A2, A1, A5) on LLM/VLM safety, out-of-distribution detection, and AI agents.

2025 – 2026 **Research Assistant** Just & Green Transportation Lab, University of Florida
Advisor: Prof. Xiang Yan
Led two projects (P8, A7) on persona-based simulation and persona-aware assessment of cycling infrastructure, and contributed to one project (W1) on multi-agent street design generation.

2025 – 2026 **Research Assistant** GLAD Lab, Arizona State University
Advisor: Prof. Xiyang Hu
Led one project (P9) on whether vision-language models can infer human engagement, and contributed to one project (W2) on vision-language model rankers.

## Industry Experience

2024 **AI Product Manager Intern** bilibili Group

2023 – 2024 **HMI UX Design Intern** BMW Group, SKY Lab

2023 **UX Design Intern** Publicis Sapient

2023 **Global Brand Design Intern** NIO Inc.

## Selected Honors & Awards

2026 **Best Paper Award** (top 1%) for "Made to Feel" (P7), IEEE VIS
2026 **Oral Presentation** (top 2.6%) for "Do Vision Language Models Understand Human Engagement in Games?" (P9), EMNLP
2026 **Special Recognition for Outstanding Reviews**, ACM CHI
2026 **Jenny Preece & Ben Shneiderman Travel Award** ($1,000), University of Maryland, College Park
2024 **Outstanding Graduate of Shanghai**, the highest honor for graduating students in Shanghai, China

## Talks & Presentations

2026 **Do Vision Language Models Understand Human Engagement in Games?**
Oral presentation, EMNLP 2026, Budapest, Hungary, October 2026

2026 **StreetDesignAI: Broadening Designer Perspectives Through Multi-Persona Evaluation of Cycling Infrastructure**
Paper presentation, DIS 2026, Singapore, June 2026

2025 **Rolling into the Future: Navigating Career Pathways with a GenAI-Powered Pool World Model**
Talk and poster, HCIL Annual Symposium, University of Maryland, College Park, May 2025
Poster, Graduate Research Appreciation Day, University of Maryland, College Park, April 2025

## Academic Service

Reviewed 30+ papers across HCI, NLP, and machine learning venues.

### Program Committees

2026 – 2027 AAAI Conference on Artificial Intelligence (AAAI)
2026 AAAI Social Impact Track

### Paper Reviews

2026 – 2027 ACM CHI Conference on Human Factors in Computing Systems (CHI) · Special Recognition for Outstanding Reviews, 2026
2026 Annual Meeting of the Association for Computational Linguistics (ACL)
2026 Conference on Empirical Methods in Natural Language Processing (EMNLP)
2026 Conference on Neural Information Processing Systems (NeurIPS)
2026 ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW)
2026 International Journal of Human–Computer Interaction (IJHCI)
2025 ACM Creativity & Cognition Conference (C&C)

### Volunteering

2026 **Student Volunteer**, ACM Designing Interactive Systems Conference (DIS)

## Teaching Experience

2025 – 2026 **Graduate Assistant**, INST201: Introduction to Information Science, College of Information, University of Maryland, College Park · Fall 2025, Spring 2026

## Skills

**Machine Learning & LLMs**: LLM/VLM evaluation and benchmarking, prompt engineering (zero-shot, theory-guided, retrieval-augmented), fine-tuning, multi-agent systems, RLHF data annotation, data preprocessing

**Programming & Tools**: Python, PyTorch, Hugging Face Transformers, LLM APIs, SQL, JavaScript, HTML/CSS, Git, LaTeX

**Human-Centered Research**: within-subjects user studies, interviews, contextual inquiry, usability testing, survey design, quantitative and qualitative (thematic) analysis

**Prototyping & Design**: React.js, Three.js, Node.js, Unity3D, Figma, Adobe Creative Suite